Moral Reasoning on the Pareto Frontier

A multi-objective view of epistemic and ethical values, and why equilibrium-style methods are well-suited to traverse it

Draft, June 2, 2026

Abstract

What counts as good moral reasoning when there is no single ground truth? We propose a multi-objective view in which a moral position — a set of commitments together with a set of principles that account for them — is evaluated against two axis families that genuinely conflict. The primary axes are *ethical*: principles carry fuzzy loadings on moral values, and we work with five — FREEDOM, SOLIDARITY, JUSTICE, WELFARE, and DIGNITY — chosen so that the central tensions of moral life (liberty vs. community, fairness vs. welfare, welfare vs. dignity) appear directly as trade-offs among axes. The secondary axes are *epistemic* measures of internal coherence, drawn from the formal account of Beisbart et al. (2021): account, systematicity, and faithfulness. Every reachable position occupies a point in this joint space, and the rational object of study is its *Pareto frontier*: the set of attainable positions for which no other position is at least as good on every axis and strictly better on at least one. We argue that this reframes the central question of moral reasoning from "which position is right?" to "which strategy reliably finds positions on the frontier?" — and that the question favours *equilibrium-style methods* that balance competing values, because without a ground truth the best a reasoner can do is settle on a defensible middle. We sketch a pilot using MoReBench dilemmas and an LLM-based reasoning system we have already prototyped.

1 Motivation: moral reasoning without a ground truth

Moral reasoning resists the model of justification that works elsewhere in epistemology: no external fact of the matter fixes which position is right. Two careful reasoners can start from different intuitions and reach different conclusions without either being simply wrong; pluralism in moral epistemology is widely accepted (Elgin, 1996; Rechnitzer, 2022). The right question is then not which position is correct but *what counts as good reasoning* in such a domain.

We argue the question has a structural answer: good moral reasoning is *multi-objective*. A position is graded along two axis families that genuinely conflict. The primary ones are *ethical* — weight on different moral values: freedom, solidarity, justice, welfare, dignity, which trade off against each other. The secondary ones are *epistemic* — the position's internal coherence: account, systematicity, faithfulness (Beisbart et al., 2021). When objectives conflict, the rational object of study is not a single optimum but a *Pareto frontier*: the set of attainable positions no other position dominates on every axis. A good reasoning strategy is one that reliably reaches it, and reaches *interior* (mixed) points — corner solutions like "maximise welfare and nothing else" are easy and uninteresting. Reflective equilibrium, with its alternating adjustment of commitments and principles, is the canonical equilibrium-style traversal of this space; it is the primary strategy we study, and we compare it to several alternatives below.

2 The moral data resource

We propose making moral reasoning empirically studyable by constructing a curated, mathematically defined *moral data resource*, then studying reasoning strategies as algorithms over it. Three ingredients combine.

Pre-enumerated inputs. Candidate case verdicts and candidate moral principles are pre-enumerated. We generate two finite pools from a dilemma source, then curate with explicit quality control: fuse near-duplicates, drop weak candidates, score each principle on quality dimensions (independent systematicity, specificity, domain-fit), and have philosophers review every retained candidate. The curated pool is meant as a reusable resource in its own right.

Pre-enumerated inferential connections. For every principle-commitment pair we LLM-generate whether the principle entails the commitment, contradicts it, or is silent on it; philosophers then review the resulting graph. The result is fixed before any reasoning strategy runs.

Pre-tagged ethical values. Each surviving principle carries a fuzzy value-loading vector distributing its weight across the five values.

Reasoning as definable movement in a defined space. Once this substrate is fixed, a reasoning strategy can be defined precisely: it is a way of moving through the space, selecting a subset of commitments and a subset of principles subject to consistency, evaluated by the multi-objective measures of Section 3. The comparison across strategies (which values does each favour? where does each converge?) becomes scientifically tractable, because *no LLM participates inside the reasoning loop* — the LLM does its work during construction, and any differences in outputs downstream reflect differences in strategies, not in the content.

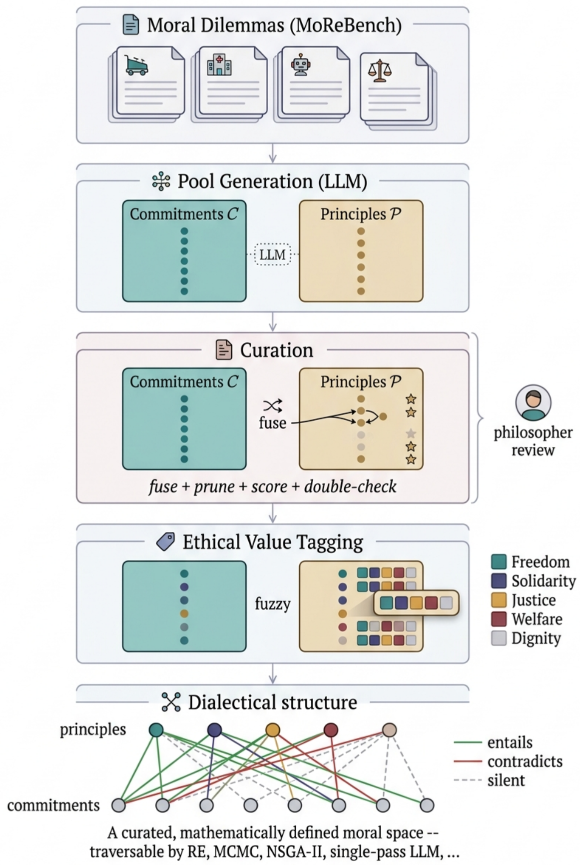

Figure 1: **Building the moral data resource.** Source dilemmas → LLM generation of commitment and principle pools → *curation* (fuse, prune, score per-principle systematicity; philosopher review) → value tagging → dialectical structure. The output is a curated, mathematically defined moral space over which any traversal method can be evaluated.

3 A formal model of the moral solution space

We adopt a formal model adapted from Beisbart et al. (2021) and the simulation infrastructure of Freivogel (2023), extended with an explicit value layer. We summarise the notation here and then state the multi-objective optimization problem it gives rise to.

Pools. Fix a finite pool of candidate moral commitments $\mathcal{S}$ — case verdicts indexed by the dilemmas in our dataset, closed under negation — a finite pool of candidate moral principles $\mathcal{P}$, and a finite set of moral values $\mathcal{V} = \{v_1, \ldots, v_K\}$. In this work $\mathcal{V} = \{\text{FREEDOM}, \text{SOLIDARITY}, \text{JUSTICE}, \text{WELFARE}, \text{DIGNITY}\}$. Note that $\mathcal{S}$ is not a domain-general pool of moral propositions; it is data-set-specific.

Value loading. Each principle carries a fuzzy value-loading vector

$$\boldsymbol{v} : \mathcal{P} \to [0,1]^K, \qquad \textstyle\sum_k \boldsymbol{v}_k(p) = 1.$$

Values are properties of principles, not of commitments, on the view that commitments are case verdicts ("in dilemma d, action a is required") whereas principles articulate *why*, and the *why* is what carries value content.

Dialectical structure. The pools are linked by a relation $\rho : \mathcal{P} \times \mathcal{S} \to \{\text{entails}, \text{contradicts}, \text{silent}\}$ that records, for every principle–commitment pair, the inferential relation between them. The triple $\tau = \langle \mathcal{S}, \mathcal{P}, \rho \rangle$ plays the role of Beisbart's dialectical structure and defines the search space.

Epistemic state. An *epistemic state* is a pair $\sigma = (C, T)$ with $C \subseteq \mathcal{S}$ minimally consistent (never contains both s and $\neg s$) and $T \subseteq \mathcal{P}$. The initial commitments $C_0 \subseteq \mathcal{S}$ are fixed before reasoning begins and serve as a faithfulness anchor.

Objectives. A state σ is evaluated by K ethical-value measures and three epistemic measures:

$$\begin{aligned} V_k(\sigma) &= \tfrac{1}{|T|} \textstyle\sum_{p \in T} \boldsymbol{v}_k(p) \in [0,1], \quad k = 1, \ldots, K \\ A(\sigma) &= \text{degree to which } T \text{ accounts for } C \in [0,1] \\ S(\sigma) &= \text{systematicity of } T \in [0,1] \\ F(\sigma \mid C_0) &= \text{faithfulness of } C \text{ to } C_0 \in [0,1] \end{aligned}$$

The value measures aggregate the loadings of the principles in T. The three coherence measures are operationalised following Beisbart et al. (2021); we refer to their paper for definitions. Together they give a *multi-objective optimization problem* over Σ, the set of admissible epistemic states:

$$\max_{\sigma \in \Sigma} \big(V_1(\sigma), \ldots, V_K(\sigma),\ A(\sigma),\ S(\sigma),\ F(\sigma \mid C_0) \big).$$

Pareto dominance and frontier. A state σ' *Pareto dominates* σ, written $\sigma \prec \sigma'$, if every objective at σ' is at least as large as at σ and at least one is strictly larger. The *Pareto frontier* of the problem is

$$\Pi(\tau, C_0) \;=\; \big\{ \sigma \in \Sigma : \nexists \sigma' \in \Sigma \text{ with } \sigma \prec \sigma' \big\}.$$

Π is the empirical object this paper proposes to study.

4 Two axis families, one frontier

The objective vector splits into two axis families: the primary $(V_1, \dots, V_K)$ is *ethical*, the secondary (A, S, F) is *epistemic*. Projecting $\Pi(\tau, C_0)$ onto each gives two lower-dimensional frontiers Π_V and Π_E, related but not collapsible: an epistemic Pareto-optimal state need not be ethical Pareto-optimal (it may achieve high coherence at a corner of value-space) and vice versa. We propose landing on $\Pi_E \cap \Pi_V$ — coherent *and* pareto-optimal across values — as the target of good moral reasoning.

Values as primary, frameworks as a secondary lens. The fine-grained value tensions that drive moral reasoning — FREEDOM against SOLIDARITY, WELFARE against DIGNITY — are clearer at the value level than at the framework level. Frameworks are bundles of values (Kantian deontology loads on DIGNITY and a particular kind of JUSTICE; utilitarianism on WELFARE and a particular kind of SOLIDARITY), and from value-tagged principles we can recover framework fits *a posteriori*. We include this analysis in the pilot.

5 Methodology sketch

Data. We use the MoReBench-Theory subset (Scale Labs, 2025): 150 scenarios with explicit framework labels, which serve as anchors for our value tagging.

Pipeline. The construction of the data resource is the load-bearing step:

(i) *Generate.* For each of ∼10 dilemmas, LLM-generate a commitment pool (∼30 candidates) and a principle pool (∼20 candidates, half value-archetypal, half bottom-up from the commitments).

(ii) *Curate.* Fuse near-duplicates, drop weak candidates, score each surviving principle on independent quality dimensions (per-principle systematicity, specificity, domain-fit). Philosopher review on every retained candidate.

(iii) *Tag.* LLM-assign a value-loading vector $\boldsymbol{v}(p)$ to each surviving principle (MoReBench-Theory framework labels seed the loadings; philosopher review confirms).

(iv) *Connect.* LLM-judge ρ for every principle-commitment pair as a one-time batch; philosopher review confirms the graph.

Traversals. On the same curated substrate we run several strategies: symbolic RE (primary), an LLM-in-loop RE prototype (already implemented on the trolley pilot, kept as a reference), a single-pass LLM judgment baseline (no equilibration), an MCMC walk over admissible states with Z as energy, multi-objective evolutionary search (NSGA-II) used to estimate Π_V and Π_E, and pure-component greedy baselines that maximise only A, only S, or only F. Each yields a distribution of $(A, S, F, V_1, \dots, V_K)$ points.

Analysis. Locate each method's landing distribution on Π_E and Π_V. Test H1 by convergence rate and Hausdorff distance of RE fixed points to NSGA-II's Π_E; test H2 by directional displacement of RE landings on Π_V relative to NSGA-II at matched epistemic cost.

6 Two frontiers, one trajectory

Figures 2 and 3 sketch the two projected frontiers and a single RE trajectory through each, with accepted and rejected moves. The epistemic trajectory is what our trolley pilot already shows: an A-step lifts A at modest S cost; a B-step expands C to admit an emerging commitment, briefly hurting A; the next A-step articulates a sharper principle that lifts both axes; the rejected moves map to dominated points off the frontier. The value trajectory is more interesting: the same alternating pattern ends not at a corner (pure freedom or pure solidarity) but at an *interior* point of Π_V — a mix neither axis alone would predict but that no other mix dominates. This is what we mean by an equilibrium in value-space, and what we suspect makes RE philosophically distinctive.

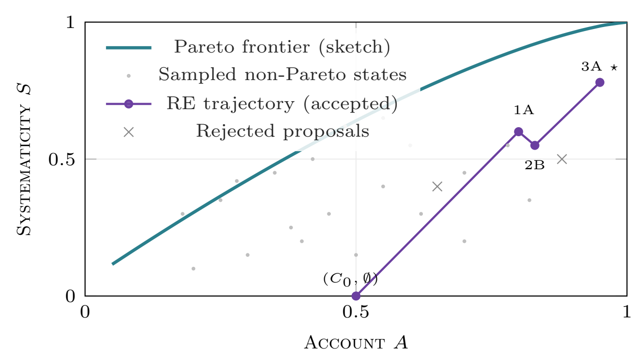

Figure 2: **Epistemic Pareto frontier.** Projection onto (A, S) with $F \approx 1$. The RE run climbs through three accepted moves and rejects two dominated proposals (grey ×). The starred final state sits at the frontier — a global trade-off optimum no other reachable state dominates.

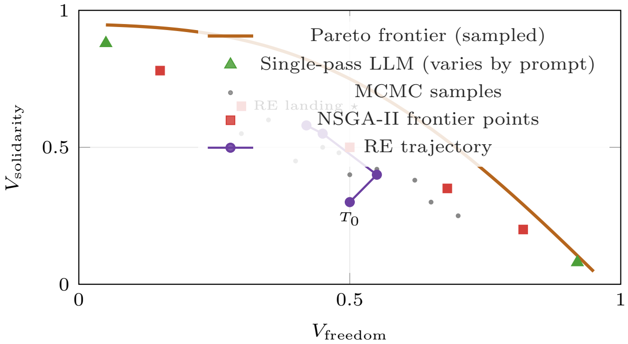

Figure 3: **Ethical-value Pareto frontier.** Projection of $V(T)$ onto $(V_{\text{freedom}}, V_{\text{solidarity}})$. Single-pass LLM judgment collapses to corners depending on prompt. NSGA-II traces the frontier. RE lands at an interior frontier point — a defensibly mixed normative profile.

7 Hypotheses

We propose two empirically testable hypotheses about reflective equilibrium. The first concerns how well RE performs as an algorithm on the epistemic side of the problem; the second concerns *where* on the ethical frontier its outputs sit.

H1 (Reliable, fast epistemic convergence). On the epistemic frontier Π_E, RE converges reliably and quickly. Across initial commitments and weight choices, the fixed point it reaches lies within ϵ of Π_E, and the number of equilibration rounds before that fixed point is small (single-digit under reasonable conditions). This is what makes RE attractive as a reasoning strategy in the first place: it produces internally coherent positions efficiently, even though it cannot guarantee a unique one.

H2 (Value-side landing region). Where RE lands on the ethical-value frontier Π_V is not arbitrary. We predict that landing points cluster in the *interior* of Π_V rather than at its corners, and that they tend to favour values whose principles admit compact statement — so welfare and justice over solidarity and dignity — though the strength of this bias should depend on the curated principle pool. If confirmed, H2 says that an internally well-motivated method of moral reasoning silently carries a substantive value bias.

8 Discussion

The framing addresses two long-standing complaints about RE without rejecting them. *Pluralism* is real but not relativism: the frontier picks out a proper subset of positions, and being on it is a substantive constraint. *Conservatism* shows up as a measurable landing-region offset toward $F = 1$, which we can now quantify rather than dismiss. The underlying philosophical question — what *is* good moral reasoning? — becomes empirically tractable: a property of traversal strategies on a measurable solution topology. The contribution we are aiming at is a measurement instrument — value-trajectory plots and Pareto-region analyses — applicable to any LLM-based or symbolic moral reasoning system.

References

Claus Beisbart, Gregor Betz, and Georg Brun. Making reflective equilibrium precise: A formal model. *Ergo*, 8 (15):441–472, 2021. doi: 10.3998/ergo.1152.

Catherine Z. Elgin. *Considered Judgment*. Princeton University Press, 1996.

Andreas Freivogel. Does reflective equilibrium help us converge? *Synthese*, 202(171), 2023. doi: 10.1007/s11229-023-04375-0.

Tanja Rechnitzer. Turning the trolley with reflective equilibrium. *Synthese*, 200(4), 2022. doi: 10.1007/s11229-022-03762-3.

Scale Labs. MoReBench: Evaluating procedural and pluralistic moral reasoning in language models. arXiv:2510.16380, `https://morebench.github.io/`, 2025.